Nahida Akther Reshmi

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EDUCATION

Jahangirnagar University, Dhaka, Bangladesh *Mar 2022 - Aug 2026*
B.Sc. in Computer Science and Engineering - GPA: 3.75/4.00 (upto 7th semester)

Birshreshtha Noor Mohammad Public College, Pilkhana, Dhaka *Aug 2018 - Feb 2020*
Higher Secondary School (HSC) - GPA: 5.0/5.0

PROJECTS

Guideline -Verifiable Multi-hop XAI for Clinical support ***(Research Project)*** *Multi-Hop XAI*

- Designed a four-layer conceptual framework for multi-hop explainable AI in clinical decision support and implemented it as System C — a 3-hop (identification → confirmation → management) reasoning pipeline verified against a guideline knowledge base spanning 6 clinical conditions, achieving 91% guideline-consistency verification, 100% rejection of out-of-scope cases, and 100% discrimination on a controlled ground-truth test across 56 real clinical notes, corroborated by a 36-rater human evaluation on clarity and trust.
- Tools: Python, LLM-based reasoning & verification (GPT-OSS 120B via OpenRouter), ChromaDB, Sentence-Transformers, RAG pipeline design

Online Art Marketplace ***(Full-Stack Project)*** *Art-Gallery*

- Full-featured art e-commerce platform with role-based access (buyer/seller/admin), artwork listing & approval workflow, cart-to-checkout order pipeline, and review/messaging/notification systems.
- Tools: Django, SQLite, Bootstrap, HTML/CSS, JS

AI-Powered Travel Planning App ***(Android Project)*** *TravelQuest*

- AI-assisted travel companion app with itinerary generation, budget estimation & destination discovery, plus a social feed for sharing travel experiences.
- Tools: Kotlin, Jetpack Compose, Firebase (Auth, Firestore), Gemini API, Cloudinary

Explainable Fake Review Detection ***(Machine Learning Project)*** *Fake-Review-Detection*

- Developed an explainable ML system for detecting fake product reviews using TF-IDF and review metadata, with optimized ensemble classifiers and SHAP/LIME-based model interpretability.
- Tools: Python, Scikit-learn, XGBoost, NLTK, SHAP, LIME, Pandas, NumPy, Gradio

Personal Expense Tracker ***(Android Application)*** *Daily Expense Tracker*

- Developed an Android expense-tracking app for recording, editing, and deleting daily expenses, with local data persistence and monthly spending visualization.
- Tools: Kotlin, Android Studio, Room Database, RecyclerView, Material Components, MPAndroidChart, Kotlin Coroutines

SKILLS

Programming Languages: Java, Python, C, C++, JavaScript
Frameworks & Platforms: Django, Android (Kotlin), Jetpack Compose
Databases: SQLite, Firebase Firestore, ChromaDB
AI/ML: Machine Learning, NLP, RAG, Explainable AI (XAI), LLM Applications, SHAP, LIME
Tools: Git, GitHub, Postman, Jupyter Notebook, Google Colab, Android Studio

REFERENCES

[1] **Dr. Md. Ezharul Islam**, Professor, CSE, Jahangirnagar University *(Research Project Supervisor)*
ezharul.islam@juniv.edu
[2] **Dr. Md. Musfique Anwar**, Professor, CSE, Jahangirnagar University *(Undergraduate Course Teacher)*
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[3] **Abir Ashab Niloy**, Graduate Teaching Assistant, University of South Florida *(Project Collaborator)*
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